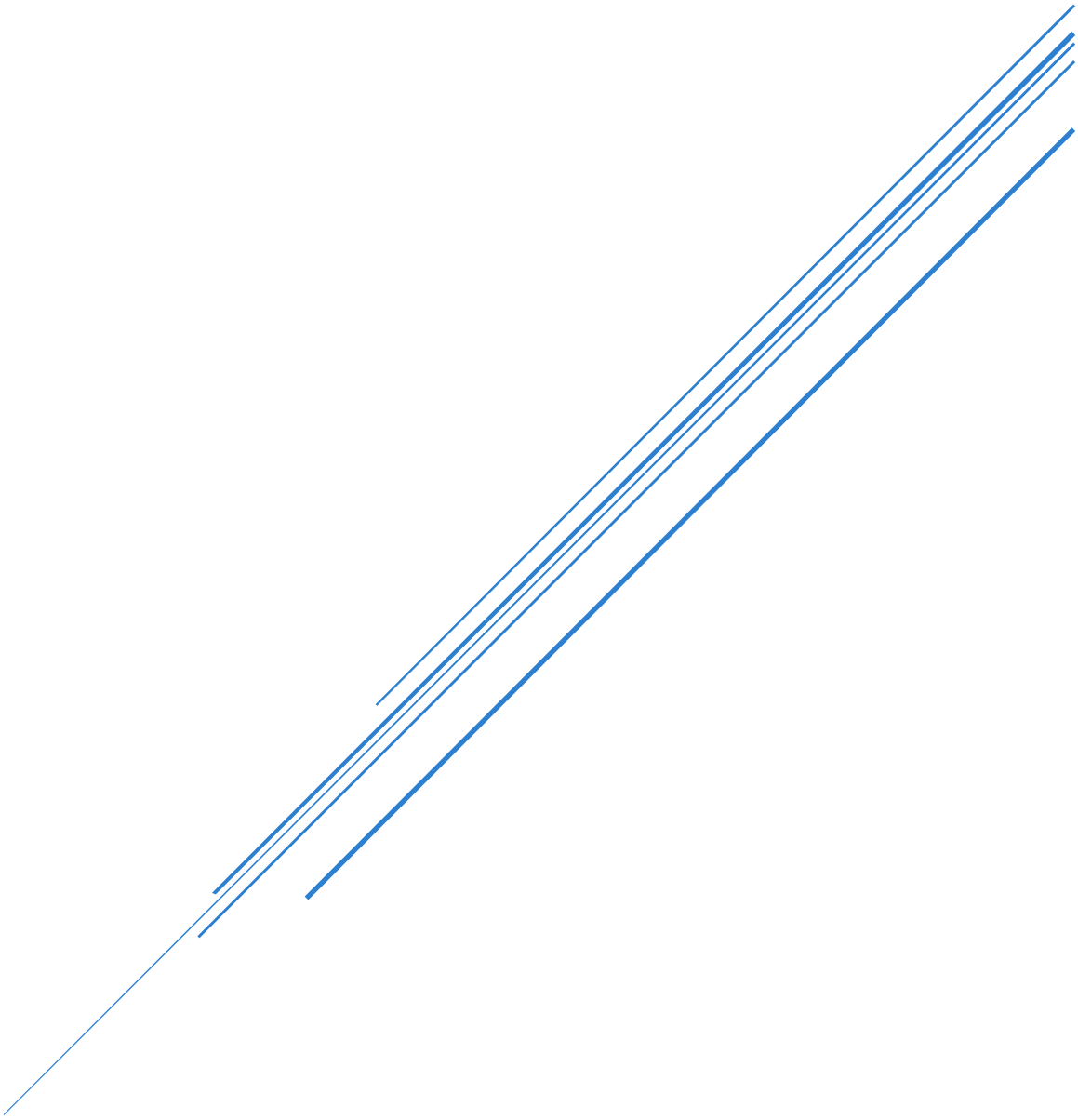
# NETWORK THREATS AND MITIGATION & PHYSICAL SECURITY AND RISK

Assignment - 4



1. **Create a table listing 5 common network threats (such as phishing, DDoS, man-in-the-middle, malware, and password attacks), and for each, write one real-world example from an app or service you use (like WhatsApp, Instagram, or Paytm).**

➢

| No. | Network Threat | Real-World Example |
|---|---|---|
| **1** | Phishing | On Instagram, attackers send fake login links through messages to steal users' passwords. |
| **2** | DDoS Attack | An attacker floods a website or service with traffic, making it unavailable to genuine users. |
| **3** | Man-in-the-Middle (MITM) | When using WhatsApp on public Wi-Fi, an attacker may try to intercept communication or steal login information. |
| **4** | Malware | A user downloads a fake Instagram APK that contains malware and can steal personal data. |
| **5** | Password Attack | Attackers try multiple passwords to access someone's Instagram or Paytm account without permission. |

2. **Simulate a brute-force attack scenario by writing a simple Python script that tries to guess a 4-digit PIN code by checking all possible combinations against a preset value.**
   **Constraint: The script should print the number of attempts it took to guess the correct PIN.**

➢

```
# Brute Force Attack Simulation


preset_pin = "2580"

attempts = 0


for pin in range(10000):

  guess = str(pin).zfill(4)

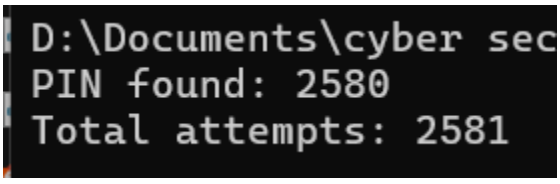
  attempts += 1


  if guess == preset_pin:

    print("PIN found:", guess)

    print("Total attempts:", attempts)

    break
```



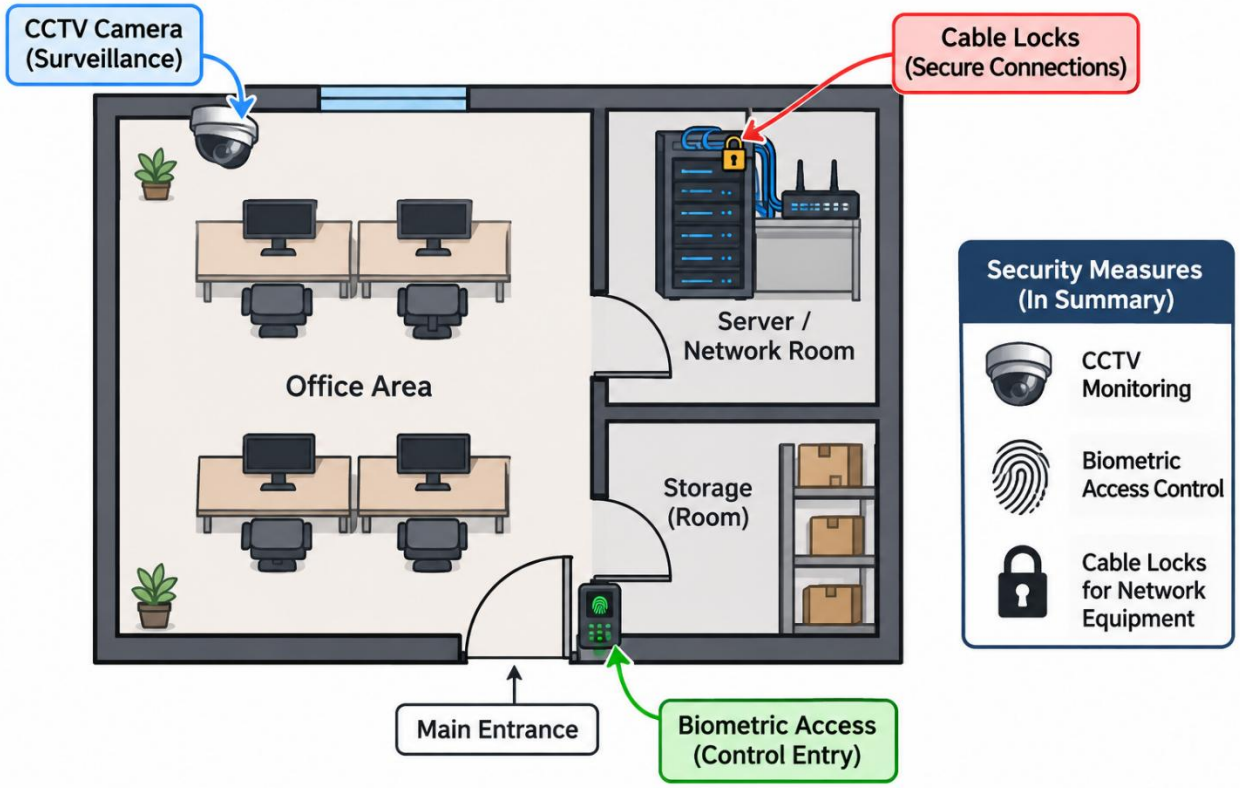
3. **List 4 physical security risks that could affect a server room hosting a popular app like Zomato or BookMyShow, and suggest one mitigation step for each risk.**

➢

| No. | Physical Security Risk | Mitigation Step |
|---|---|---|
| **1** | Unauthorized Access: An unknown person enters the server room and steals or damages equipment. | Use biometric access control or ID cards. |
| **2** | Fire: Fire can damage servers and destroy important data. | Install fire alarms and automatic fire suppression systems. |
| **3** | Power Failure: A sudden power cut can shut down servers and affect apps like Zomato. | Use UPS and backup generators. |
| **4** | Overheating: High temperatures can cause servers to malfunction or shut down. | Install proper cooling systems and temperature monitoring. |

4. **Draw a simple floor plan (hand-drawn or digital) of a small office and mark at least 3 physical security measures you would implement to protect network equipment (such as CCTV, biometric access, or cable locks).**

➢



5. **Use ChatGPT or a similar AI tool to generate a checklist of 7 actions a network admin should take to mitigate risks from both network threats and physical security breaches, then review and edit the checklist to make it specific for a college computer lab environment.**

➢

| No. | Checklist Action | How to Implement in a College Lab |
|---|---|---|
| **1** | Install Firewall | Configure a firewall to block unauthorized network traffic and suspicious connections. |
| **2** | Use Strong Passwords | Give each student a separate login ID and password for lab computers. |
| **3** | Install Antivirus | Keep antivirus updated on all lab computers to detect and remove malware. |
| **4** | Install CCTV Cameras | Monitor lab entrances, computer areas, and network equipment. |
| **5** | Restrict Physical Access | Allow only authorized faculty and IT staff to enter the server and network equipment room. |
| **6** | Update Software Regularly | Install security updates for Windows, applications, routers, and switches. |
| **7** | Take Regular Backups | Keep backups of important student records, lab files, and college data in a secure location. |

**Review and Editing**

After reviewing the AI-generated checklist, I made it specific to a college computer lab by:

- Including separate student login accounts.
- Adding CCTV monitoring for lab equipment.
- Restricting access to network devices and server rooms.
- Including regular backups of student and college data.